

What ADE Could Do

Product Capability Paper for This Monorepo

Shipped spine → H-track bets → user-facing futures

Companion: *Ideal ADE: Applying Human-Agent Market Findings*

Basis: Orch G0–G4 · Effort B0–B4 · DEC-A-014 · live .ade/

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Type: Product capability paper (this codebase)

Audience: Maintainers deciding H-track bets

Date stance: 2026-07 — *this* ADE, not a generic agent IDE

Abstract

ADE already ships more harness spine than most “AI IDE” demos: Suggest / Apply / Automate, leases, Isolate worktrees, Verify G0–G5, Continuity capsules, SpendGuard reserve/reconcile, eng-goals, skills/rules/profiles packs, and a Desktop control plane. What it **could** do next is close the gap to the ideal ADE: honest dollars, contract-before-Apply, slot Orchestrator truth, model profiles, richer lease UX, and action envelopes in the trace — without waiting on Zed ACP.

This paper states **what ADE can become as a product**, ranked by leverage against habits the market study named, with shipped evidence and concrete next bets.

1. Who ADE is today (one paragraph)

ADE is a **local harness / multi-host agent OS**: Desktop is the control plane; crates under `agents / workflow / core` are the brain; `.ade/` on disk is shared identity. Autonomy maps Suggest $\approx$ Propose, Apply $\approx$ Act under leases, Automate $\approx$ Act + required verify. Orch G0–G4 and Effort B0–B4 are done. VSCodium is retired. Zed ACP is scaffold (`ade acp --probe` only). Critical path is harness depth (DEC-A-014), not editor wrapping.

2. What ADE already can do (shipped leverage)

2.1 Run agents under a turn law

- Every turn ends `completed | failed | cancelled` in the feed.
- Budgets, Continue / Fix&retry, thrift Continuity — Effort B0–B4.
- **Could mean for users:** fewer blank-retry death spirals; failures are actionable.

2.2 Separate plan from mutate

- Suggest queues PLAN / tasks; Apply claims one under leases.
- Isolate Apply $\rightarrow$ `.ade/worktrees/{task}`.
- **Could mean:** parallel work without peer-chat chaos; one writer per path when dogfooded.

2.3 Gate risk with ToolEffect + rules

- Authority classifies effects; `.ade/rules` (secrets, money, build-risk, generated-deny, ...) always-on.
- Skills (verify-ladder, handoff-continuity, accidental-data-loss-prevention, ...) activate on demand.
- **Could mean:** safety without dumping every procedure into every prompt.

2.4 Verify outside the generator

- VerifyRunner G0–G5; Automate requires verify-on-complete.
- Gold harness (~51 tasks) via `ade eval --gold`.
- **Could mean:** self-certify is structurally harder than in chat-only tools.

2.5 Continuity instead of paste thrash

- Capsules (ade.handoff/v1), host next_safe_command, resume thrift.
- **Could mean:** long missions survive budget stops without dumping the thread.

2.6 Meter spend (basic)

- Reserve / reconcile, session + daily caps, Trust Spend UI, integer micros money helpers.
- **Could mean:** hard stops exist — honesty of the \$ number is the gap (see §4).

2.7 Eng-goals as first-class objects

- Create / list / active / run under .ade/goals/.
- Desktop Role strip + intent terminalization (G0–G3).
- **Could mean:** contract spine exists; hard gate before Apply is the unlock.

2.8 Desktop as control plane

- Home composer, Guided/Power/Dev, Terminal, Monaco editor + sensitive paths, Workspaces, Browser, Trust, Atlas, Recipes, BYOK + model picker.
- **Could mean:** full agent OS UX without claiming “we are an IDE.”

3. Gap to ideal (honest scorecard)

Ideal principle	This ADE	Gap
Envelope per side effect	ToolEffect + leases + PLAN ≈ partial	Persist full envelope in turn / Continuity
Contract before Apply	Goals exist; Apply not hard-gated	Eng-goal / clarify gate
Slot hierarchy product	Role strip + claim-one	Slot Orchestrator + heartbeats
Honest spend	Caps + ledger	Rates / reserve / cache honesty (H1)
Model profiles	Guidance packs; ModelSelector stub	ade.model-profile/v1 + router (H3)
Lease conflict UX	Enforcement + strip helper	Actionable CTAs (H4)
Risk-tiered HITL	Autonomy dial + PLAN approve	Policy object for secrets/infra/migrate
Editor eyes	ACP probe only	Optional after H1–H3 (DEC-A-014)
Continuity	Strong	Defaults / dogfood polish
Verify	Strong	Contract-linked cmds; race gold (H5)

4. What ADE could do next (product bets)

Ranked by leverage for *this* repo vs market habits.

Bet 1 — SpendGuard honesty (H1)

Could do: Trust Spend that matches invoice class — real rates (no silent \$0), sane reserve, cache fields when known, clear used / reserved / remaining.

Kills: "I thought I had budget" + opaque \$/outcome.

Depends on: existing ledger + money helpers.

Not: Mission Control charts.

Bet 2 — Contract gate before Apply

Could do: Refuse Act-class tools until active eng-goal has AC + OOS + verify pointer — or ≤ 3 clarify chips resolve.

Kills: underspec → expensive mutate → redo (largest human waste in the study).

Depends on: existing goal store + next-actions chips.

Ship shape: Suggest freely; Apply blocked with CTA "define goal."

Bet 3 — Slot Orchestrator truth (H2)

Could do: Bind Suggest=Planner, Apply=Worker, verify=Judge to task claim + heartbeats; make dual-writer impossible even when UI is misused.

Kills: peer multi-agent tax / two workers on one path.

Depends on: leases + tasks registries already on disk.

Defer: Mission Control until slots are real.

Bet 4 — Model profiles + router (H3)

Could do: `ade.model-profile/v1` — autonomy ceiling, tool mask, effort floor, spend ceiling, slot eligibility; router replaces `ModelSelector` → `None`.

Kills: model roulette mid-task; wrong-model-for-role spend.

Depends on: `ModelPicker` + `SessionProfile` stubs.

User-visible: "why this model" chip on turn start.

Bet 5 — Lease conflict product UX (H4 slice)

Could do: Blocked Apply explains who holds what; CTAs: wait · Isolate · rotate lease agent.

Kills: silent conflict / red badge with no path.

Depends on: `AgentSessionStrip` + registry.

Bet 6 — Authorized action envelopes in the trace

Could do: Persist effect · paths · lease · autonomy · plan/goal id per tool grant into turn events and Continuity.

Kills: unauditable side effects; weak replay.

Unlocks: regulated / enterprise packs later without redesign.

Bet 7 — Harness gold races (H5)

Could do: Expand gold with dual-writer, wrong-slot, budget stop, spend-cap halt, Isolate merge regressions.

Kills: "works in demo, fails under load."

Depends on: existing 51-task gold + `ade eval --gold`.

Bet 8 — Reproduce-first + evidence-graded research skills

Could do: On-demand skills that force repro / evidence before patch or claim.

Kills: sycophantic “looks good” research and coding thrash.

Cheap: markdown + activate path already exists.

Bet 9 — Risk-tiered HITL policy

Could do: Secrets / infra / migrate / publish force Propose or human confirm even under Automate; log waivers.

Kills: review habituation on high-blast paths.

Depends on: ToolEffect + build-risk / secrets rules.

Bet 10 — Zed ACP soft shell (Z?, after H1–H3)

Could do: Full ACP stdio mapping turns → Zed Agent Panel; Open-in-Zed for coding eyes.

Does not: redefine product identity; DEC-A-014 keeps this optional.

Fork ladder: only if ACP+Desktop cannot close written chrome gaps.

5. Narrative product futures (what users experience)

Future A — “Contracted Apply”

User states intent → ADE opens eng-goal → ≤3 clarify → Suggest plans → Apply claims leased task → verify → Continuity if budget bites. **Feeling:** less thrash, more finished work per dollar.

Future B — “Honest Trust”

User sees spend that matches provider reality; caps halt; Continue resumes thrift without paste. **Feeling:** agent OS you can run overnight without invoice shock.

Future C — “Slot crew”

Planner proposes; Worker mutates one claim; Verifier grades; conflicts offer Isolate. **Feeling:** multi-agent that doesn’t step on itself.

Future D — “Profiled models”

Cheap / strong / tool-light profiles bound to slots and spend ceilings. **Feeling:** amplification without roulette.

Future E — “Optional Zed eyes”

Same .ade/ identity; Desktop control plane; Zed for code browsing when ACP is ready. **Feeling:** harness OS with better eyes — not “we forked an editor.”

6. What ADE should not try to be (near-term)

Temptation	Why not (for this ADE)
Cursor/Zed clone as identity	DNA + DEC-A-014: harness first

Flat swarm of peer agents	CooperBench-style peer tax; leases already reject this
Enterprise Mission Control first	Needs slot truth + envelopes first
Vertical SaaS (sales/support) as core	Packs later; coding harness is the wedge
Blocking all progress on ACP	Scaffold only; must not gate H1–H3

7. Capability map — today → could

TODAY	COULD (H-track)
Suggest / Apply / Automate	→ + contract gate + risk HITL
Leases + Isolate	→ + conflict CTAs + heartbeats
Verify G0–G5 + Automate	→ + eng-goal-linked cmds + race gold
Continuity + thrift	→ + default polish
Spend caps + ledger	→ + invoice-honest SpendGuard
Goals + Role strip	→ + hard gate before Act tools
Guidance packs	→ + model profiles + router
ToolEffect + PLAN	→ + persisted action envelopes
ACP --probe	→ + soft shell (optional, later)
Gold ~51	→ + multi-agent / budget races

8. Recommended sequencing (this repo)

Per DEC-A-014: pick **one** primary of H1 or H2, then the other; H3 rides with H2; H4/H5 continuous; Z after dogfood.

Now H1 SpendGuard honesty OR H2 Slot Orchestrator
Next Contract-before-Apply (pairs with either)
Then H3 model profiles + router
Parallel H4 lease UX • H5 gold races • envelope-in-trace • skills
Later Zed ACP M2 soft shell
Held Mission Control • SSO/SCIM • hard fork

Highest combined ROI: H1 + contract gate — dollars and redo-loops are the two market wounds ADE is already positioned to close.

9. Success criteria for “ADE could do this”

Ship a dogfoodable story:

1. **Cannot Apply without a contract** (or explicit waive logged).
2. **Cannot Automate-complete without verify.**
3. **Cannot dual-write** a leased path (slot + lease).
4. **Spend UI within tolerance of invoice** for a known model.
5. **Continuity resume** after budget stop without paste.
6. **Gold green** including at least one dual-writer / wrong-slot case.

When those six hold, ADE is not “an IDE with chat” — it is a **harness OS** that applies the market study for real.

10. Conclusion

What ADE could do: become the local agent OS that makes the market study’s winning pattern unavoidable — contracts, envelopes, leases, verify, honest spend, Continuity, profiled models — while keeping Desktop as control plane and Zed as optional eyes.

What ADE already is: halfway there on spine; short on honesty, gates, and slot productization.

What to build: H1 or H2 first, contract gate close behind, profiles next, ACP last among equals.

Appendix A — Shipped evidence (paths)

Capability	Path hint
Turn / autonomy / authority	crates/agents/src/{turn,autonomy,authority}.rs
Skills / goals / spend / handoff	crates/agents/src/{skills,goal,spend,handoff}.rs
Leases / verify / tasks	crates/workflow/src/{parallel,verify,tasks}.rs
Guidance profiles	crates/core/src/guidance.rs · .ade/profiles/
Rules / skills on disk	.ade/rules/ · .ade/skills/
Desktop control plane	apps/desktop/
ACP scaffold	crates/acp/
Decisions	docs/decisions/DEC-A-010 ... 014
DNA	AGENTS.md

Appendix B — Relation to papers

Paper	Role
<i>Agents Meet Humans</i>	Agnostic market / habit evidence
<i>Ideal ADE From Market Study</i>	Normative architecture
<i>What ADE Could Do</i> (this)	Instantiation on this monorepo
<i>ADE Tokenomics</i>	I/O · context · usage strategies (not \$)
Harness / amplification brief	Harness · MAS · verify · \$ metering adjacent